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#-----
# Vivado v2025.2 (64-bit)
# SW Build 6299465 on Fri Nov 14 19:35:11 GMT 2025
# IP Build 6300035 on Fri Nov 14 10:48:45 MST 2025
# SharedData Build 6298862 on Thu Nov 13 04:50:51 MST 2025
# Start of session at: Sat Jul 25 12:14:25 2026
# Process ID           : 6936
# Current directory   :
D:/Vivado/CORDIC_Processor/CORDIC_Processor.runs/synth_1
# Command line        : vivado.exe -log cordic_top.vds -product
Vivado -mode batch -messageDb vivado.pb -notrace -source
cordic_top.tcl
# Log file            :
D:/Vivado/CORDIC_Processor/CORDIC_Processor.runs/synth_1/cordic_top
.vds
# Journal file        :
D:/Vivado/CORDIC_Processor/CORDIC_Processor.runs/synth_1\vivado.jou
# Running On          : AK_LAPTOP
# Platform             : Windows Server 2016 or Windows 10
# Operating System     : 26200
# Processor Detail     : Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz
# CPU Frequency        : 2496 MHz
# CPU Physical cores   : 4
# CPU Logical cores    : 8
# Host memory          : 8419 MB
# Swap memory          : 15032 MB
# Total Virtual        : 23451 MB
# Available Virtual    : 9110 MB
#-----
source cordic_top.tcl -notrace
create_project: Time (s): cpu = 00:00:16 ; elapsed = 00:00:13 .
Memory (MB): peak = 524.465 ; gain = 217.871
Command: read_checkpoint -auto_incremental -incremental
D:/Vivado/CORDIC_Processor/CORDIC_Processor.srcs/utils_1/imports/sy
nth_1/cordic_top.dcp
INFO: [Vivado 12-5825] Read reference checkpoint from
D:/Vivado/CORDIC_Processor/CORDIC_Processor.srcs/utils_1/imports/sy
```

nth_1/cordic_top.dcp for incremental synthesis

INFO: [Vivado 12-7989] Please ensure there are no constraint changes

Command: synth_design -top cordic_top -part xc7a15tcp236-2

Starting synth_design

Attempting to get a license for feature 'Synthesis' and/or device 'xc7a15t'

INFO: [Common 17-349] Got license for feature 'Synthesis' and/or device 'xc7a15t'

INFO: [Device 21-403] Loading part xc7a15tcp236-2

INFO: [Designutils 20-5440] No compile time benefit to using incremental synthesis; A full resynthesis will be run

INFO: [Designutils 20-4379] Flow is switching to default flow due to incremental criteria not met. If you would like to alter this behaviour and have the flow terminate instead, please set the following parameter config_implementation

{autoIncr.Synth.RejectBehavior Terminate}

INFO: [Synth 8-7079] Multithreading enabled for synth_design using a maximum of 2 processes.

INFO: [Synth 8-7078] Launching helper process for spawning children vivado processes

INFO: [Synth 8-7075] Helper process launched with PID 31688

Starting RTL Elaboration : Time (s): cpu = 00:00:09 ; elapsed = 00:00:11 . Memory (MB): peak = 1260.098 ; gain = 543.000

INFO: [Synth 8-6157] synthesizing module 'cordic_top'

[D:/Projects/CORDIC/codes/cordic_top.v:1]

INFO: [Synth 8-6157] synthesizing module 'control_fsm'

[D:/Projects/CORDIC/codes/control_fsm.v:1]

Parameter ITERATION bound to: 16 - type: integer

INFO: [Synth 8-155] case statement is not full and has no default
[D:/Projects/CORDIC/codes/control_fsm.v:101]

INFO: [Synth 8-6155] done synthesizing module 'control_fsm' (0#1)
[D:/Projects/CORDIC/codes/control_fsm.v:1]

```
INFO: [Synth 8-6157] synthesizing module 'atan_rom'
[D:/Projects/CORDIC/codes/atan_rom.v:1]
    Parameter WIDTH bound to: 16 - type: integer
    Parameter ADDR_WIDTH bound to: 4 - type: integer
INFO: [Synth 8-6155] done synthesizing module 'atan_rom' (0#1)
[D:/Projects/CORDIC/codes/atan_rom.v:1]
INFO: [Synth 8-6157] synthesizing module 'datapath'
[D:/Projects/CORDIC/codes/datapath.v:1]
    Parameter WIDTH bound to: 16 - type: integer
INFO: [Synth 8-6157] synthesizing module 'gen_reg'
[D:/Projects/CORDIC/codes/gen_reg.v:1]
    Parameter WIDTH bound to: 16 - type: integer
INFO: [Synth 8-6155] done synthesizing module 'gen_reg' (0#1)
[D:/Projects/CORDIC/codes/gen_reg.v:1]
INFO: [Synth 8-6157] synthesizing module 'add_sub'
[D:/Projects/CORDIC/codes/add_sub.v:1]
    Parameter WIDTH bound to: 16 - type: integer
INFO: [Synth 8-6155] done synthesizing module 'add_sub' (0#1)
[D:/Projects/CORDIC/codes/add_sub.v:1]
INFO: [Synth 8-6157] synthesizing module 'gain_comp'
[D:/Projects/CORDIC/codes/gain_comp.v:1]
    Parameter WIDTH bound to: 16 - type: integer
INFO: [Synth 8-6155] done synthesizing module 'gain_comp' (0#1)
[D:/Projects/CORDIC/codes/gain_comp.v:1]
INFO: [Synth 8-6155] done synthesizing module 'datapath' (0#1)
[D:/Projects/CORDIC/codes/datapath.v:1]
WARNING: [Synth 8-689] width (4) of port connection 'iter' does not
match port width (3) of module 'datapath'
[D:/Projects/CORDIC/codes/cordic_top.v:48]
INFO: [Synth 8-6155] done synthesizing module 'cordic_top' (0#1)
[D:/Projects/CORDIC/codes/cordic_top.v:1]
-----
-----
Finished RTL Elaboration : Time (s): cpu = 00:00:12 ; elapsed =
00:00:14 . Memory (MB): peak = 1377.535 ; gain = 660.438
-----
-----
```


Start Handling Custom Attributes

Finished Handling Custom Attributes : Time (s): cpu = 00:00:12 ;
elapsed = 00:00:14 . Memory (MB): peak = 1377.535 ; gain = 660.438

Finished RTL Optimization Phase 1 : Time (s): cpu = 00:00:12 ;
elapsed = 00:00:14 . Memory (MB): peak = 1377.535 ; gain = 660.438

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.008 . Memory (MB): peak = 1377.535 ; gain = 0.000
INFO: [Project 1-570] Preparing netlist for logic optimization

Processing XDC Constraints

Initializing timing engine

Parsing XDC File

[D:/Vivado/CORDIC_Processor/CORDIC_Processor.srcs/constrs_1/new/clock_constraint.xdc]

Finished Parsing XDC File

[D:/Vivado/CORDIC_Processor/CORDIC_Processor.srcs/constrs_1/new/clock_constraint.xdc]

Completed Processing XDC Constraints

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00 . Memory (MB): peak = 1443.660 ; gain = 0.000

INFO: [Project 1-111] Unisim Transformation Summary:

No Unisim elements were transformed.

Constraint Validation Runtime : Time (s): cpu = 00:00:00 ; elapsed

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= 00:00:00.033 . Memory (MB): peak = 1443.660 ; gain = 0.000
INFO: [Designutils 20-5440] No compile time benefit to using
incremental synthesis; A full resynthesis will be run
INFO: [Designutils 20-4379] Flow is switching to default flow due
to incremental criteria not met. If you would like to alter this
behaviour and have the flow terminate instead, please set the
following parameter config_implementation
{autoIncr.Synth.RejectBehavior Terminate}
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Finished Constraint Validation : Time (s): cpu = 00:00:26 ; elapsed
= 00:00:30 . Memory (MB): peak = 1443.660 ; gain = 726.562
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Start Loading Part and Timing Information
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Loading part: xc7a15tcpg236-2
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-----
Finished Loading Part and Timing Information : Time (s): cpu =
00:00:26 ; elapsed = 00:00:30 . Memory (MB): peak = 1443.660 ; gain
= 726.562
-----
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-----
-----
Start Applying 'set_property' XDC Constraints
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-----
-----
Finished applying 'set_property' XDC Constraints : Time (s): cpu =
00:00:26 ; elapsed = 00:00:30 . Memory (MB): peak = 1443.660 ; gain
= 726.562
```


INFO: [Synth 8-802] inferred FSM for state register 'state_reg' in
module 'control_fsm'

State	New Encoding
Previous Encoding	

IDLE	00001
000	
LOAD	00010
001	
ITERATE	00100
010	
GAIN	01000
011	
DONE	10000
100	

INFO: [Synth 8-3354] encoded FSM with state register 'state_reg'
using encoding 'one-hot' in module 'control_fsm'

Finished RTL Optimization Phase 2 : Time (s): cpu = 00:00:26 ;
elapsed = 00:00:30 . Memory (MB): peak = 1443.660 ; gain = 726.562

Start RTL Component Statistics

Detailed RTL Component Info :
+---Adders :

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        3 Input      16 Bit      Adders := 3
        2 Input      4 Bit      Adders := 1
+---Registers :
            16 Bit      Registers := 3
            4 Bit      Registers := 1
+---Muxes :
        2 Input      16 Bit      Muxes := 8
        5 Input      5 Bit      Muxes := 1
        2 Input      5 Bit      Muxes := 3
        2 Input      1 Bit      Muxes := 4
        5 Input      1 Bit      Muxes := 1
-----
-----
Finished RTL Component Statistics
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-----
Start Part Resource Summary
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-----
Part Resources:
DSPs: 45 (col length:60)
BRAMs: 50 (col length: RAMB18 60 RAMB36 30)
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-----
Finished Part Resource Summary
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-----
Start Cross Boundary and Area Optimization
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-----
DSP Report: Generating DSP DP/gain/x_multi, operation Mode is:
A2*(B:0x136f).
DSP Report: register DP/X_reg/q_reg is absorbed into DSP
```

DP/gain/x_multi.
DSP Report: operator DP/gain/x_multi is absorbed into DSP
DP/gain/x_multi.
DSP Report: Generating DSP DP/gain/y_multi, operation Mode is:
A2*(B:0x136f).
DSP Report: register DP/Y_reg/q_reg is absorbed into DSP
DP/gain/y_multi.
DSP Report: operator DP/gain/y_multi is absorbed into DSP
DP/gain/y_multi.

Finished Cross Boundary and Area Optimization : Time (s): cpu =
00:00:31 ; elapsed = 00:00:35 . Memory (MB): peak = 1443.660 ; gain
= 726.562

Sort Area is DP/gain/x_multi_2 : 0 0 : 949 949 : Used 1 time 0
Sort Area is DP/gain/y_multi_0 : 0 0 : 949 949 : Used 1 time 0

Start ROM, RAM, DSP, Shift Register and Retiming Reporting

DSP: Preliminary Mapping Report (see note below. The ' indicates
corresponding REG is set)

+-----+-----+-----+-----+-----+-----+-----+									
-----+-----+-----+-----+-----+-----+-----+									
Module Name DSP Mapping A Size B Size C Size D Size									
P Size AREG BREG CREG DREG ADREG MREG PREG									
+-----+-----+-----+-----+-----+-----+-----+									
-----+-----+-----+-----+-----+-----+-----+									
cordic_top A2*(B:0x136f) 16 14 - -									
30 1 0 - - - 0 0									
cordic_top A2*(B:0x136f) 16 14 - -									
30 1 0 - - - 0 0									
+-----+-----+-----+-----+-----+-----+-----+									

-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
Note: The table above is a preliminary report that shows the DSPs
inferred at the current stage of the synthesis flow. Some DSP may
be reimplemented as non DSP primitives later in the synthesis flow.
Multiple instantiated DSPs are reported only once.

Finished ROM, RAM, DSP, Shift Register and Retiming Reporting

Start Applying XDC Timing Constraints

Finished Applying XDC Timing Constraints : Time (s): cpu = 00:00:46
; elapsed = 00:00:51 . Memory (MB): peak = 1618.773 ; gain =
901.676

Start Timing Optimization

Finished Timing Optimization : Time (s): cpu = 00:00:47 ; elapsed =
00:00:52 . Memory (MB): peak = 1659.016 ; gain = 941.918

Start Technology Mapping

Finished Technology Mapping : Time (s): cpu = 00:00:48 ; elapsed =
00:00:53 . Memory (MB): peak = 1659.016 ; gain = 941.918

Start IO Insertion

Start Flattening Before IO Insertion

Finished Flattening Before IO Insertion

Start Final Netlist Cleanup

Finished Final Netlist Cleanup

Finished IO Insertion : Time (s): cpu = 00:00:55 ; elapsed =
00:01:00 . Memory (MB): peak = 1883.090 ; gain = 1165.992

Start Renaming Generated Instances

Finished Renaming Generated Instances : Time (s): cpu = 00:00:55 ;
elapsed = 00:01:00 . Memory (MB): peak = 1883.090 ; gain = 1165.992

Start Rebuilding User Hierarchy

Finished Rebuilding User Hierarchy : Time (s): cpu = 00:00:55 ;
elapsed = 00:01:00 . Memory (MB): peak = 1883.090 ; gain = 1165.992

Start Renaming Generated Ports

Finished Renaming Generated Ports : Time (s): cpu = 00:00:55 ;
elapsed = 00:01:00 . Memory (MB): peak = 1883.090 ; gain = 1165.992


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Start Handling Custom Attributes
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-----
Finished Handling Custom Attributes : Time (s): cpu = 00:00:55 ;
elapsed = 00:01:00 . Memory (MB): peak = 1883.090 ; gain = 1165.992
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-----
Start Renaming Generated Nets
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-----
-----
Finished Renaming Generated Nets : Time (s): cpu = 00:00:55 ;
elapsed = 00:01:00 . Memory (MB): peak = 1883.090 ; gain = 1165.992
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-----
-----
Start Writing Synthesis Report
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-----
DSP Final Report (the ' indicates corresponding REG is set)
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
|Module Name | DSP Mapping | A Size | B Size | C Size | D Size | P
Size | AREG | BREG | CREG | DREG | ADREG | MREG | PREG |
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
|cordic_top  | A'*B       | 30      | 13      | -      | -      | 30
      | 1      | 0      | -      | -      | -      | 0      | 0      |
|cordic_top  | A'*B       | 30      | 13      | -      | -      | 30
      | 1      | 0      | -      | -      | -      | 0      | 0      |
```

Report BlackBoxes:

BlackBox name	Instances

Report Cell Usage:

	Cell	Count
1	BUFG	1
2	CARRY4	12
3	DSP48E1	2
4	LUT1	1
5	LUT2	4
6	LUT3	23
7	LUT4	9
8	LUT5	66
9	LUT6	40
10	FDRE	56
11	FDSE	1
12	IBUF	52
13	OBUF	49

Finished Writing Synthesis Report : Time (s): cpu = 00:00:55 ;
elapsed = 00:01:00 . Memory (MB): peak = 1883.090 ; gain = 1165.992

Synthesis finished with 0 errors, 0 critical warnings and 0
warnings.

Synthesis Optimization Runtime : Time (s): cpu = 00:00:40 ; elapsed

= 00:00:57 . Memory (MB): peak = 1883.090 ; gain = 1099.867
Synthesis Optimization Complete : Time (s): cpu = 00:00:55 ;
elapsed = 00:01:01 . Memory (MB): peak = 1883.090 ; gain = 1165.992
INFO: [Project 1-571] Translating synthesized netlist
Netlist sorting complete. Time (s): cpu = 00:00:01 ; elapsed =
00:00:00.010 . Memory (MB): peak = 1887.516 ; gain = 0.000
INFO: [Netlist 29-17] Analyzing 14 Unisim elements for replacement
INFO: [Netlist 29-28] Unisim Transformation completed in 0 CPU
seconds
INFO: [Project 1-570] Preparing netlist for logic optimization
INFO: [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).
Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00 . Memory (MB): peak = 1889.188 ; gain = 0.000
INFO: [Project 1-111] Unisim Transformation Summary:
No Unisim elements were transformed.

Synth Design complete | Checksum: a0632b5b
INFO: [Common 17-83] Releasing license: Synthesis
37 Infos, 1 Warnings, 0 Critical Warnings and 0 Errors encountered.
synth_design completed successfully
synth_design: Time (s): cpu = 00:01:02 ; elapsed = 00:01:27 .
Memory (MB): peak = 1889.188 ; gain = 1358.988
Write ShapeDB Complete: Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.011 . Memory (MB): peak = 1889.188 ; gain = 0.000
INFO: [Common 17-1381] The checkpoint
'D:/Vivado/CORDIC_Processor/CORDIC_Processor.runs/synth_1/cordic_to
p.dcp' has been generated.
INFO: [Vivado 12-24828] Executing command : report_utilization
-file cordic_top_utilization_synth.rpt -pb
cordic_top_utilization_synth.pb
INFO: [Common 17-206] Exiting Vivado at Sat Jul 25 12:16:15 2026...